

# Deva Anand

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## Education

### University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

**Coursework:** Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

### Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

## Skills

### Languages

Python, C++, Java, Go, JavaScript, TypeScript, SQL, Bash

### Data & Performance

Apache Spark / PySpark, Spark UI, distributed-data processing, performance optimization, Linux profiling, data pipelines

### Backend & Reliability

REST APIs, monitoring & observability, log monitoring, incident response, production debugging, automated & integration testing, multi-tenant systems

### Cloud & DevOps

AWS (S3, KMS), GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

### Databases

PostgreSQL, MongoDB, Redis, SQLite

## Experience

### Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability and security for **multi-tenant NHS eTriage systems** (Node.js, MongoDB) serving **8+ UK clients** and processing **10,000+ monthly triage submissions** integrated with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Monitored production logs and system behavior across releases and client integrations for operational **observability**, investigated incidents, and shipped fixes for tenant-specific and integration issues, improving reliability and stability of the data-facing backend.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive clinical workflows against regressions; enforced secure-by-default patterns across tenant isolation and backend access paths.

### Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.

## Projects

### Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, SQL, Python | 1.4B-row dataset

- Cut end-to-end **data-processing** runtime roughly **25%** (17 to 19 min down to 14.3 min) on a **1.4B-row** dataset by replacing shuffle joins with broadcast joins, enabling adaptive query execution and skew-join handling, and tuning shuffle partitions, to scale efficiently and cut compute cost.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks, reducing analytics tail-latency ratio (P99/P50) from **2.14x** to **1.74x** via partition/column pruning, filter pushdown, and precomputed fields.

### cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf\_ai\_research\_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an edge-to-cloud assistant on Cloudflare Workers that fetches sources and streams results to the client over **WebSocket** in real time, prototyped end-to-end with embedded storage.
- Designed durable workflow boundaries so multi-step tasks survive process restarts and tool failures; state persists server-side in D1 without an external database.

### AgenticSearch - Provenance-First Data Pipeline

github.com/Deva-1903/ciir\_agentic\_search

Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage data pipeline (typed retrieval planning, extraction with deterministic parsers before LLM fallback, output verification) that returns cell-level provenance for every value.
- Hardened the service with **150+ automated tests**, provider routing/fallback, and pipeline instrumentation for reliable, debuggable behavior, so changes merge and deploy with confidence.

## Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.